

Roadmap to Junior Data Analyst / Data Scientist (15 hours per week)

This roadmap is designed for candidates with basic programming knowledge who want to land a Junior Data Analyst or Junior Data Scientist role or internship.

Language: English

Study time: 15 hours per week

Duration: ~6 months

STAGE 1 – Python for Data Analysis (Weeks 1–3)

Skills:

- Pandas, NumPy
- Data cleaning
- Basic visualization

Resources:

- Kaggle Learn – Python: <https://www.kaggle.com/learn/python>
- Kaggle Learn – Pandas: <https://www.kaggle.com/learn/pandas>
- FreeCodeCamp Python for Data Analysis: <https://www.youtube.com/watch?v=r-uOLxNrNk8>

Project:

- Exploratory analysis on any Kaggle dataset

STAGE 2 – Data Analysis & Visualization (Weeks 4–6)

Skills:

- EDA
- Grouping, aggregations
- Matplotlib, Seaborn

Resources:

- Kaggle Data Visualization: <https://www.kaggle.com/learn/data-visualization>
- Pandas Documentation: <https://pandas.pydata.org/docs/>
- Fundamentals of Data Visualization (free): <https://clauswilke.com/dataviz/>

Project:

- Full EDA report with insights and charts

STAGE 3 – SQL for Data Analysis (Weeks 7–10)

Skills:

- SELECT, WHERE, GROUP BY
- JOINS
- Subqueries, CTEs

Resources:

- Kaggle SQL: <https://www.kaggle.com/learn/intro-to-sql>
- SQLBolt: <https://sqlbolt.com/>
- Mode SQL Tutorial: <https://mode.com/sql-tutorial/>
- FreeCodeCamp SQL course: <https://www.youtube.com/watch?v=HXV3zeQKqGY>

Project:

- SQL case study answering business questions

STAGE 4 – Data Storytelling (Weeks 11–12)

Skills:

- Chart selection
- Communication of insights

Resources:

- Storytelling with Data (summary): <https://www.storytellingwithdata.com/>
- Tableau Public: <https://public.tableau.com/>

Project:

- One-page analytical report or dashboard

STAGE 5 – Power BI (Weeks 13–15)

Skills:

- Power Query
- DAX basics
- Interactive dashboards

Resources:

- Microsoft Learn Power BI: <https://learn.microsoft.com/en-us/training/powerplatform/power-bi/>
- Power BI Full Course: <https://www.youtube.com/watch?v=AGrl-H87pRU>
- Guy in a Cube: <https://www.youtube.com/@GuyinaCube>

Project:

- Interactive Power BI dashboard

STAGE 6 – Statistics (Weeks 16–17)

Skills:

- Descriptive statistics
- Probability
- Hypothesis testing

Resources:

- Khan Academy Statistics: <https://www.khanacademy.org/math/statistics-probability>
- StatQuest: <https://www.youtube.com/@statquest>

Project:

- Statistical analysis with conclusions

STAGE 7 – Machine Learning (Weeks 18–21)

Skills:

- Regression
- Classification
- Model evaluation

Resources:

- Kaggle ML: <https://www.kaggle.com/learn/intro-to-machine-learning>
- Google ML Crash Course: <https://developers.google.com/machine-learning/crash-course>
- Andrew Ng ML: <https://www.coursera.org/learn/machine-learning>

Project:

- Titanic or House Prices ML project

STAGE 8 – Portfolio & Job Prep (Weeks 22–26)

Skills:

- Resume
- GitHub portfolio
- Interview prep

Resources:

- StrataScratch: <https://www.stratascratch.com/>
- Data Interview Questions: <https://www.datacamp.com/blog/data-analyst-interview-questions>
- GitHub Pages: <https://pages.github.com/>

Portfolio:

- 3–5 projects (SQL, EDA, Power BI, ML)

FINAL GOAL:

Be job-ready for Junior Data Analyst / Data Scientist or internship positions.